

WebGPU-Accelerated Multimodal Content Generation for Immersive XR Presentations

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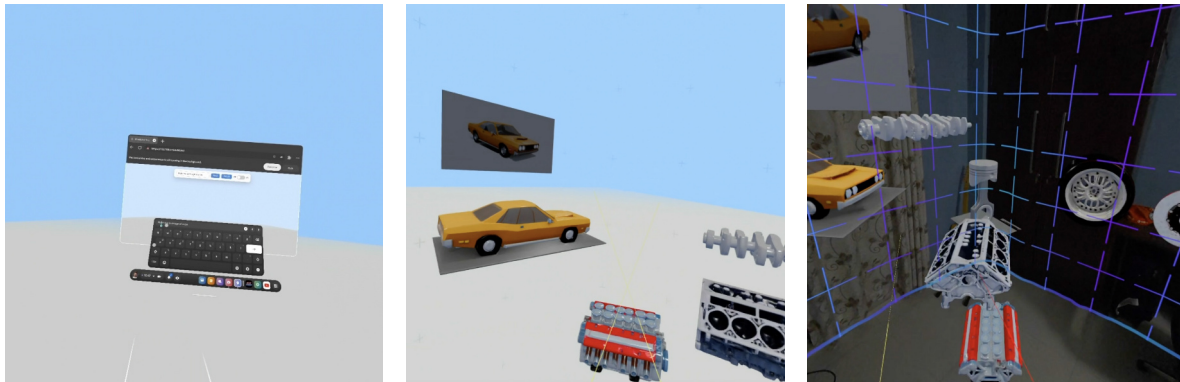


Figure 1: Cross-reality XR presentation workflow. From left to right: authoring a natural-language prompt in virtual reality (VR), generated 3D objects visualized and interacted with in VR, and the same content deployed in augmented reality (AR) within physical space.

ABSTRACT

We present a novel system that reimagines how users create and deliver presentations within extended reality (XR). Beyond static 2D slides, our approach enables presenters to generate images or 3D objects dynamically using natural language input, supporting immersive storytelling and interactive Q&A in real time. The system leverages a modular architecture combining voice or text input, a WebGPU-enabled frontend for local AI inference, and a backend MCP-based pipeline for text-to-image-to-3D generation. The prototype demonstrates how presenters can create museum-like or spatially rich presentations, respond to unexpected questions with new visualizations, and explore spontaneous content generation within an XR environment.

Index Terms: Extended Reality, WebGPU, Generative AI, Text-to-3D, Immersive Presentation.

1 INTRODUCTION AND MOTIVATION

Traditional slide-based presentations often constrain the presenter's creativity and engagement, particularly in immersive or educational contexts. Extended Reality (XR) offers spatial and interactive capabilities, but most XR tools still rely on pre-authored content.

Our system enables presenters to create immersive XR presentations that go beyond static slides. Presenters can describe concepts or objects via voice or text and immediately generate corresponding 2D images or 3D objects within a shared XR scene. This allows for:

- **Museum-like presentations:** Objects and exhibits can be arranged spatially, providing an engaging narrative for viewers.
- **Dynamic Q&A support:** If a presenter is asked a question for which no prepared material exists, they can generate relevant images or 3D object in real time.
- **Spontaneous storytelling:** New objects or scenes can be created in response to audience interaction.
- **Exploratory learning:** Users can manipulate, annotate, and compare generated objects in real time, enhancing understanding.

By combining real-time generative AI with XR interaction, the system demonstrates how live content creation can enrich presentations, improve engagement, and extend the presenter's expressive capabilities.

2 RELATED WORK

Our approach builds on generative workflows for metaverse asset creation [3], which introduced a three-tier AI pipeline for 3D asset generation. While prior works focused on asset creation and optimization, we extend these ideas to live, interactive XR presentations where content is generated on demand.

Unlike approaches relying entirely on cloud GPUs, our system explores local inference on WebGPU, enabling partial on-device computation with lightweight models, particularly Whisper for speech-to-text. This design provides a proof-of-concept to run the whole AI pipeline locally, provided the required hardware.

3 DESIGN GOALS AND CHALLENGES

Our system emphasizes three primary design goals: **responsiveness**, **portability**, and **interactivity**.

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- **Responsiveness:** Minimizing latency between input and generated content is essential to maintain presentation flow.
- **Portability:** Support for WebGPU-based inference on standalone or browser-based XR devices.
- **Interactivity:** Users can manipulate generated 3D objects, explore spatial arrangements, and adapt content dynamically.

Challenges include synchronizing user input with real-time AI inference, optimizing model sizes for limited GPU memory, and ensuring consistent performance within browser-based XR environments.

4 SYSTEM OVERVIEW

The system integrates a WebXR/WebGPU frontend with a modular backend built around a FastMCP pipeline.

4.1 Frontend (WebXR and WebGPU)

The frontend supports:

- Text input through an immersive user interface.
- Local speech-to-text inference using Whisper on WebGPU.
- Voice-based command interaction for object manipulation, including scaling, rotation, and deletion.
- Real-time rendering of generated images and objects in XR.

4.2 Backend: Unified Generation Endpoint

The backend exposes a single endpoint, `/generate`, which serves as the entry point for all frontend requests. Instead of providing separate endpoints for image or 3D generation, the system relies on an AI agent to interpret the user's input and decide the appropriate generation pipeline. When a request arrives, the AI agent determines whether the user intends to:

- Generate a 2D image from text (text $\rightarrow$ image).
- Produce a 3D object from a textual description (text $\rightarrow$ image $\rightarrow$ 3D object).

Once the pipeline is chosen, the AI agent invokes the corresponding tool on the FastMCP server:

- `generate_image_from_text`: Uses Z-Image for text-to-image generation [1, 2, 7].
- `generate_object_from_image`: Uses Hunyuan3D-2 to convert an image to a 3D object (GLB, base64-encoded) [4, 5].
- `generate_object_from_text`: A wrapped pipeline that first generates an image with Z-Image and then uses Hunyuan3D-2 to produce a 3D object. Direct text-to-object conversion is not reliable, so this sequential approach ensures better quality [4, 5, 7].

All outputs are returned as base64-encoded assets, which simplifies integration with the XR frontend and allows seamless streaming of images or 3D objects into the scene. By centralizing generation through a single endpoint, the system maintains a clean API while supporting flexible, context-aware content creation.

4.3 User Interaction Flow

The flow begins with user speech or text input, which is structured into a prompt and sent through the generation pipeline:

1. User speech or text input is processed locally.
2. Requests are forwarded to the backend for asset generation.
3. Generated assets are streamed into the XR environment.
4. Users interact with objects using standard WebXR gestures or voice commands.

This workflow enables adaptive, live presentations that respond dynamically to audience interaction.

5 DISCUSSION AND FUTURE VISION

The prototype demonstrates that interactive AI-driven 3D content generation during XR presentations is already feasible. While current standalone headsets constrain full on-device inference, emerging PC-class AR/VR systems could significantly reduce these limitations.

Future systems such as the Steam Frame [6] may execute the entire text-to-image-to-3D pipeline locally or via low-latency PC offloading with its 6GHz wireless dongle, improving privacy, responsiveness, and visual fidelity. Beyond performance gains, this shift redefines the presenter's role from static designer to live content creator.

Beyond technical feasibility, this work also raises broader questions about human-AI collaboration in immersive settings. Real-time generative systems may shift the presenter's role from static designer to live creator, demanding new interaction paradigms that balance automation with user control. Future work will investigate optimization techniques for quantized and distilled generative models, enabling efficient on-device training or fine-tuning. Ethical considerations include ensuring that generated 3D content aligns with user intent, maintains fidelity, and preserves intellectual property integrity. Embedding transparent AI attribution could enhance trust and accountability in generative XR experiences.

6 CONCLUSION

We presented an immersive XR presentation system that integrates WebGPU-based inference, a modular FastMCP backend, and real-time 3D rendering. The system enables dynamic, museum-like presentations, supports real-time Q&A, and demonstrates new possibilities for AI-assisted storytelling in XR. Ongoing work will focus on fully local pipelines and richer interaction techniques.

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